Prabhavit Gupta

GenAI Integration Architect: OpenAI Ecosystem, Secure APIs & LLM Orchestration

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Summary

GenAI Integration Architect with 3 years of commercial experience designing, securing, and deploying enterprise-grade LLM integrations, conversational platforms, and autonomous agent systems. Deep expertise in the OpenAI and frontier model ecosystems: REST APIs, Python SDKs, Server-Sent Events (SSE) streaming, function calling, rate limit governance, and cost/token optimization. Proven record in securing enterprise API boundaries (OAuth2, JWT, mTLS, proxies), orchestrating multi-agent reasoning with LangGraph and LangChain, and implementing automated observability and tracing. Based in Venice, Italy with permanent EU work authorization (Permesso di Soggiorno UE a Tempo Indeterminato, immediate start for full remote).

Experience

FlowSpace.co • *Lead GenAI Integration Architect & Developer (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Architected and deployed production GenAI integration pipelines connecting OpenAI and Anthropic model APIs to core enterprise services, supporting 4,500+ active users with 99.9% uptime.
- Implemented real-time Server-Sent Events (SSE) streaming, dynamic function calling, and structured JSON output validation layers in Python.
- Designed resilient API gateway patterns with adaptive rate limiting, exponential backoff retries, and token usage optimization to prevent service outages.
- Secured LLM-to-backend communication using OAuth2, JWT verification, and encrypted webhook listeners.
- Engineered high-precision RAG pipelines featuring document ingestion, hybrid semantic search, metadata filtering, and reranking to eliminate hallucinations.
- Built automated evaluation and tracing harnesses using LangSmith and Docker across AWS and GCP environments.
- *Tools: Python 3, OpenAI API, LangGraph, LangChain, RAG, pgvector, OAuth2, JWT, FastAPI, Docker, Git*

Urlink • *AI Systems Integration Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Developed automated document extraction and classification pipelines connecting external partner APIs with centralized database stores.
- Implemented semantic chunking and structured prompt validation layers in Python, ensuring data integrity across high-volume ingestion streams.
- Partnered with engineering and product teams to translate functional business requirements into scalable, secure AI architecture specifications.
- *Tools: Python, REST APIs, OAuth2, PostgreSQL, Supabase, Git, Linux*

Independent Technical Consultant • *GenAI Solutions Architect*
Sep 2023 - Mar 2024 • Venice, Italy

- Designed custom conversational AI agents, document search engines, and workflow automation microservices for European SMB clients.
- Implemented function calling, webhook listeners, and secure API integrations to connect LLMs with existing enterprise software stacks.
- Authored technical architecture runbooks, prompt engineering guidelines, and deployment specifications for client engineering teams.
- *Tools: Python, LangChain, OpenAI API, REST APIs, Linux, Git*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Built automated webhook consumers and data integration workers in Python for SaaS client onboarding flows.
- *Tools: Python, REST APIs, Sendgrid*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** Permesso di Soggiorno UE per Soggiornanti di Lungo Periodo (Italy), unrestricted work authorization across the EU (CCNL, Contrato Indefinido or Partita IVA). No visa sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (working beginner).
- **Availability:** Available immediately for full-time remote roles in Spain / Europe.

Core Skills

- **OpenAI Ecosystem & LLMs:** OpenAI API, Python SDK, Server-Sent Events (SSE) Streaming, Function / Tool Calling, Structured Outputs, Rate Limit Optimization.
- **LLM Orchestration & Agents:** LangGraph, LangChain, LlamaIndex, Autonomous Agents, Prompt Engineering, Evaluation Guardrails.
- **API Architecture & Security:** RESTful APIs, OAuth2, JWT, mTLS, Webhooks, API Gateway Management, Proxy Configurations.
- **Backend & Cloud Systems:** Python 3, FastAPI, PostgreSQL, Docker, AWS, Azure, Linux, Git.
- **RAG & Vector Search:** Retrieval-Augmented Generation, Semantic Search, Hybrid Search, Vector Databases (pgvector, Chroma).